

四阶 Runge-Kutta 方法求常微分数值解报告



代码:

```
#include <stdio.h>
#include <math.h>

double f(double x, double y);

double RK4(double (*f)(double, double), double a, double b, double y0, int N, FILE *outfile) {
    double h = (b - a) / N;
    double x_0 = a;
    double y_0 = y0;
    double x, y;

    for (int i = 0; i < N; ++i) {
        double K1 = f(x_0, y_0);
        double K2 = f(x_0 + h / 2, y_0 + h * K1 / 2);
        double K3 = f(x_0 + h / 2, y_0 + h * K2 / 2);
        double K4 = f(x_0 + h, y_0 + h * K3);
        y = y_0 + h * (K1 + 2 * K2 + 2 * K3 + K4) / 6;
        x_0 += h;
        y_0 = y;
        fprintf(outfile, "%lf %lf\n", x_0, y_0);
    }
}

int main() {
    FILE *outfile=fopen("out.txt","w");
    int N;
    double a,b,y0;
    a=0.0;b=2.0;
    y0=0.5;
    N=10;
    RK4(f,a,b,y0,N,outfile);
    fclose(outfile);
}

double f(double x, double y) {
    return y - pow(x, 2) + 1;
}
```

结果:

 out - 记事本

文件(F) 编辑(E) 格式(O) 查看(V)

```
0.200000 0.829293
0.400000 1.214076
0.600000 1.648922
0.800000 2.127203
1.000000 2.640823
1.200000 3.179894
1.400000 3.732340
1.600000 4.283409
1.800000 4.815086
2.000000 5.305363
```

Matlab 验证:

```
clear all;
clc;
options=odeset('Reltol',1e-4,'Refine',1,'Maxstep',0.2);
[x,y]=ode45(@(x,y)y-x^2+1,[0 2],0.5,options)
ANSWER(:,1)=x(:,1);
ANSWER(:,2)=y(:,1);
```

12x2 double			
AN...	1	2	
1	0	0.5000	
2	0.0423	0.5647	
3	0.2423	0.9062	
4	0.4423	1.3020	
5	0.6423	1.7466	
6	0.8423	2.2331	
7	1.0423	2.7530	
8	1.2423	3.2960	
9	1.4423	3.8495	
10	1.6423	4.3981	
11	1.8423	4.9231	
12	2	5.3055	